

Developer Mock Test Results

Test ID:	e8fa0514-8b3c-43...	Student ID:	N/A
Test Type:	Developer Assessment	Date:	2026-03-04 17:56:25
Overall Score:	19/25 (76.0%)	Performance:	Good
Warnings:	0/3	Status:	Completed

Section-wise Performance

Section	Score	Percentage	Status
Aptitude	9/10	90.0%	Pass
MCQ/Theory	8/10	80.0%	Pass
Coding	2/5	40.0%	Needs Work

Detailed Question Review

APTITUDE SECTION (9/10 - 90.0%)

- Q1. A store buys a product for \$50 and sells it for \$60. What is the profit percentage?

Your Answer: 20%

Well done!
- Q2. The average of 8 numbers is 12. If one of the numbers is 15, what is the sum of the remaining 7 numbers?

Your Answer: 80

Excellent! Right answer.
- Q3. If 5 men can build a wall in 10 days, how many men are needed to build the same wall in 5 days?

Your Answer: 10

Excellent! Right answer.
- Q4. If 18 workers can build a house in 12 days, how many workers will be needed to build the same house in 10 days?

Your Answer: 22 workers

Correct! Well done.
- Q5. If it is true that all cats are animals and all dogs are animals, what can be concluded about cats and dogs?

Your Answer: They are both animals

Excellent! Right answer.

■ Q6. A mixture of paint and water is in the ratio 2:3. If 2 cups of paint are added to the mixture, what is the new ratio of paint to water?

Your Answer: 4:3

■ Excellent! Right answer.

■ Q7. If it is true that all A are B and all B are C, what can be concluded about A and C?

Your Answer: All A are C

■ Excellent! Right answer.

■ Q8. If 6 workers can build a house in 10 days, how many workers will be needed to build the same house in 8 days?

Your Answer: 8 workers

■ Correct! Well done.

■ Q9. If it is true that all books are written and all articles are written, what can be concluded about books and articles?

Your Answer: They are the same type of writing

Correct Answer: They are both written

■ The correct answer is "They are both written" because the given statements "all books are written" and "all articles are written" only confirm that both books and articles share the common characteristic of being written, but do not imply they are the same type of writing. The user's answer is likely incorrect because they mistakenly assumed that sharing a common characteristic (being written) means books and articles are the same type of writing, which is not necessarily true. This conclusion is based on logical reasoning rather than a mathematical calculation.

■ Q10. A mixture is made of 2 parts oil and 5 parts water. What is the ratio of oil to the total mixture?

Your Answer: 2/7

■ Well done!

■ MCQ SECTION (8/10 - 80.0%)

■ Q1. What is a deadlock in Java?

Your Answer: A situation where two or more threads are blocked indefinitely

■ Correct! Well done.

■ Q2. How are variables declared in Java?

Your Answer: Using the "var" keyword

Correct Answer: Using the data type

■ The correct answer is "Using the data type" because in Java, variables are declared by specifying the data type of the variable, followed by the variable name. For example, `int x` declares an integer variable named `x`. The user's answer is incorrect because Java does not use the "var" keyword to declare variables, unlike some other programming languages, such as JavaScript or C#.

■ Q3. What is the next step after creating a new Java file?

Your Answer: Write your code

■ Correct! Well done.

■ Q4. What is an example of a data type in Java?

Your Answer: Int

■ Well done!

■ Q5. What is a characteristic of Java?

Your Answer: Platform-independent

■ Excellent! Right answer.

■ Q6. What is starvation in Java?

Your Answer: A situation where a thread is able to access a shared resource

Correct Answer: A situation where a thread is unable to access a shared resource

■ The correct answer is A) A situation where a thread is unable to access a shared resource, because starvation in Java refers to a scenario where a thread is consistently denied access to a shared resource, often due to other threads holding onto it for extended periods. The user's answer, B) A situation where a thread is able to access a shared resource, is incorrect because it describes the opposite of starvation. The user's likely mistake is misunderstanding the concept of starvation, which is a type of thread scheduling problem that can lead to thread starvation, where a thread is perpetually unable to gain access to the resource it needs.

■ Q7. What is an example of a control structure in Java?

Your Answer: If-else statement

■ Well done!

■ Q8. What is an abstract class in Java?

Your Answer:

A class that cannot be instantiated but can be inherited

■ Correct! Well done.

■ Q9. What is an example of a method in Java?

Your Answer: All of the above

■ Correct! Well done.

■ Q10. What is a tool used for Java development?

Your Answer: All of the above

■ Excellent! Right answer.

■ CODING SECTION (2/5 - 40.0%)

■ Q1. Write a Java program that reads a matrix of integers from stdin and finds the sum of its elements.

****Input Format:**** - Line 1: The number of rows in ...

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();
        int cols = sc.nextInt();

        int sum = 0;

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                sum += sc.nextInt();
            }
        }
    }
}
```

```

        sc.close();

        System.out.println(sum);
    }
}

```

Correct Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int rows = scanner.nextInt();
        int cols = scanner.nextInt();
        scanner.nextLine(); // Consume newline left-over

        int sum = 0;
        for (int i = 0; i < rows; i++) {
            String[] line = scanner.nextLine().split(" ");
            for (String num : line) {
                sum += scanner.nextInt();
                scanner.nextLine(); // Consume newline left-over
            }
        }
        System.out.println(sum);
        scanner.close();
    }
}

```

■ The student's code failed 3 out of 5 test cases, indicating that the logic for calculating the sum of matrix elements is likely correct, but there might be an issue with handling input or edge cases. The code does not account for potential exceptions, such as invalid input or matrix dimensions. Without more specific error messages, it's difficult to pinpoint the exact issue, but the code's failure to pass all test cases suggests that it may not handle all possible input scenarios.

■ **Q2. Write a Java program that reads an array of integers from stdin and searches for a specified value.**

****Input Format:** - Line 1: The number of element...**

Your Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int target = sc.nextInt();
        sc.close();

        int index = -1;

        for (int i = 0; i < n; i++) {
            if (arr[i] == target) {
                index = i;
                break;
            }
        }
    }
}

```

```
        System.out.println(index);
    }
}
```

■ Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and its solution. Great job on achieving a perfect score of 5/5, your code is accepted and meets all requirements.

■ **Q3. Write a Java program that reads a stack of integers from stdin and finds the sum of its elements.**

****Input Format:** - Line 1: The number of elements ...**

Your Answer:

```
import java.util.Scanner;
import java.util.Stack;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        Stack<Integer> stack = new Stack<>();

        for (int i = 0; i < n; i++) {
            stack.push(sc.nextInt());
        }

        sc.close();

        int sum = 0;

        while (!stack.isEmpty()) {
            sum += stack.pop();
        }

        System.out.println(sum);
    }
}
```

■ Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and its requirements. Great job on achieving a perfect score of 5/5, your solution is correct and effective.

■ **Q4. Write a Java program that reads a matrix of integers from stdin and finds the minimum element.**

****Input Format:** - Line 1: The number of rows in the ...**

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();
        int cols = sc.nextInt();

        int min = Integer.MAX_VALUE;

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                int num = sc.nextInt();
                if (num < min) {
                    min = num;
                }
            }
        }
    }
}
```

```

        }
    }

    sc.close();

    System.out.println(min);
}
}

```

Correct Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int rows = scanner.nextInt();
        int cols = scanner.nextInt();
        scanner.nextLine(); // Consume newline left-over

        int min = Integer.MAX_VALUE;
        for (int i = 0; i < rows; i++) {
            String[] line = scanner.nextLine().split(" ");
            for (String num : line) {
                int numInt = Integer.parseInt(num);
                if (numInt < min) {
                    min = numInt;
                }
            }
        }

        System.out.println(min);
        scanner.close();
    }
}

```

■ The code failed 1 out of 5 test cases, indicating that it may not be handling edge cases or specific input formats correctly. The issue is likely related to the input or output format, as the code appears to correctly find the minimum element in the matrix. The exact cause of the failure cannot be determined without more information about the specific test case that failed.

■ **Q5. Write a Java program that reads a square matrix of integers from stdin and finds the sum of its diagonal elements. ****Input Format:**** - Line 1: The si...**

Your Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int sum = 0;

        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                int value = sc.nextInt();
                if (i == j) { // Main diagonal condition
                    sum += value;
                }
            }
        }

        sc.close();
    }
}

```

```
        System.out.println(sum);
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int size = scanner.nextInt();
        int[][] matrix = new int[size][size];
        for (int i = 0; i < size; i++) {
            for (int j = 0; j < size; j++) {
                matrix[i][j] = scanner.nextInt();
            }
        }
        int sum = 0;
        for (int i = 0; i < size; i++) {
            sum += matrix[i][i];
        }
        System.out.println(sum);
        scanner.close();
    }
}
```

■ The code failed to pass one test case, indicating that the issue may be related to the input or output handling. The provided code seems to correctly calculate the sum of the diagonal elements, but the missing semicolon at the end of the `System.out.println(sum)` statement could potentially cause issues. However, since the test results do not explicitly mention a syntax error, it is likely that the issue lies elsewhere, such as with the input handling or a specific test case that the code is not accounting for.

■ Recommendations

Areas that need improvement:

- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.